NexusOS
AI Business Operating System

Part 03.0
Linear Issue WAL-14

Engineering Review Reconstruction

Project	NexusOS — AI Business Operating System
Part	Part 03.0
Linear issue	WAL-14
Report title	Engineering Review Reconstruction
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Purpose	Standalone Linear attachment (manual upload)
Source report(s)	docs/reports/Part-03.0/Part-03.0-Engineering-Review-Reconstruction.md

This PDF is a presentation export of repository engineering report content. Source report files in the repository were not modified.

Evidence Statement

This report is a retrospective engineering reconstruction produced after implementation. Its contents are derived exclusively from repository evidence. It is not a contemporaneous engineering document. No engineering events are fabricated. Where repository evidence is unavailable, that limitation is explicitly documented.

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Report Content

Content below is copied from the source repository report(s) without alteration.

Part 03.0 — Engineering Review Reconstruction

Linear: WAL-14 **Reconstruction date:** 2026-07-19 **Nature:** Reconstruction only — **not** a contemporaneous independent review transcript

Evidence Statement

This report is a retrospective engineering reconstruction produced after implementation.

Its contents are derived exclusively from repository evidence.

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No engineering events are fabricated.

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Objective (as recorded in Linear)

Conduct an engineering review of the database architecture and implementation against enterprise standards before approving Part 03.0.

What repository evidence can support

The following can be verified today without inventing findings:

- **Architecture ↔ implementation consistency**

Models, migration `0a961c2ea2cf`, seed, and repository tenant filter align with `docs/product/database.md` standards (UUID v7, denormalized `company_id`, soft-delete/append-only split, RBAC shape, ON DELETE policies as coded).

- **Automated tenancy review aid**

Five isolation tests at landing encode the critical multi-tenant invariants.

- **Linear workflow gate**

WAL-14 is Done; remediation (WAL-15), documentation (WAL-16), and final verification (WAL-17) followed.

- **Later independent scrutiny of the same foundation**

Part 03.1's independent review (`Part-03.1-Independent-Engineering-Review.md`, WAL-26) examined this schema/auth foundation and found additional integrity/security gaps (e.g. missing composite tenant FKs, FK drift). Those findings belong to Part 03.1 chronology; they prove 03.0 closeout was a baseline approval, not a claim of permanent perfection.

What repository evidence cannot support

No contemporaneous Part 03.0 engineering-review finding register exists in the repository:

- No finding IDs
- No severity table from a 03.0 review session
- No signed review report dated at implementation time
- No meeting notes or checklist artifacts under `docs/reports/Part-03.0/` prior to this reconstruction

Therefore this report **does not invent** review findings. It reconstructs only what the code, docs, tests, git, and Linear workflow prove.

Linear historical scope text

WAL-14 historically listed review scope themes: architecture, schema, code quality, security, performance, scalability, best practices. Those labels describe intended review coverage; they are not evidence that specific defects were written down.

Outcome (reconstructed)

Part 03.0 passed a Linear engineering-review gate sufficient to proceed to remediation closeout, documentation, and final verification of the enterprise data-model package. Finding-level contemporaneous detail is unavailable.

Evidence limitations (shared)

- Part 03.0 code primarily landed in git commit `838db64` (2026-07-09) under the commit message “chore: validate and stabilize backend foundation”, bundled with broader foundation scaffolding. There is no separate contemporaneous commit uniquely titled “Part 03.0”.
- Linear issues WAL-7…WAL-17 were created on 2026-07-18 as a historical execution hierarchy after the schema already existed in the repository.
- No contemporaneous Part-03.0 independent-review finding IDs, meeting notes, or signed review PDF exist in the repository (unlike Part 03.1’s report pack).
- Structural integrity remediations (composite cross-tenant FKs, soft-delete-safe unique indexes, missing `created_by`/`updated_by` FKs) are **Part 03.1** evidence (`f521d38`, revisions `3a2aec52f6d2` / `217e90d00cc0`) and are not attributed as silent Part 03.0 remediations.